

Problem 1)

Solution: The first thing to do is draw all the forces on the two masses. These are shown in Fig. 3.2. The forces on M_2 are gravity and the tension. The forces on M_1 are gravity, friction, the tension, and the normal force. Note that the friction force points down the plane, because we are assuming that M_1 moves up the plane.

Having drawn all the forces, we can now write down all the $F = ma$ equations. When dealing with M_1 , we could break things up into horizontal and vertical components, but it is much cleaner to use the components parallel and perpendicular to the plane.⁵ These two components of $\mathbf{F} = \mathbf{ma}$, along with the vertical $F = ma$ equation for M_2 , give

$$\begin{aligned} T - f - M_1 g \sin \theta &= M_1 a, \\ N - M_1 g \cos \theta &= 0, \\ M_2 g - T &= M_2 a, \end{aligned} \quad (3.4)$$

where we have used the fact that the two masses accelerate at the same rate (and we have defined the positive direction for M_2 to be downward). We have also used the fact that the tension is the same at both ends of the string, because otherwise there would be a net force on some part of the string which would then undergo infinite acceleration, because it is massless.

There are four unknowns in Eq. (3.4) (namely T , a , N , and f), but only three equations. Fortunately, we have a fourth equation: $f = \mu N$, because we are assuming that M_1 is in fact moving, so we can use the expression for kinetic friction. Using this in the second equation above gives $f = \mu M_1 g \cos \theta$. The first equation then becomes $T - \mu M_1 g \cos \theta - M_1 g \sin \theta = M_1 a$. Adding this to the third equation leaves us with only a , so we find

$$a = \frac{g(M_2 - \mu M_1 \cos \theta - M_1 \sin \theta)}{M_1 + M_2} \implies T = \frac{M_1 M_2 g(1 + \mu \cos \theta + \sin \theta)}{M_1 + M_2}. \quad (3.5)$$

Note that in order for M_1 to in fact accelerate upward (that is, $a > 0$), we must have $M_2 > M_1(\mu \cos \theta + \sin \theta)$. This is clear from looking at the forces along the plane.

REMARK: If we instead assume that M_1 is sufficiently large so that it slides down the plane, then the friction force points up the plane, and we find (as you can check),

$$a = \frac{g(M_2 + \mu M_1 \cos \theta - M_1 \sin \theta)}{M_1 + M_2}, \quad \text{and} \quad T = \frac{M_1 M_2 g(1 - \mu \cos \theta + \sin \theta)}{M_1 + M_2}. \quad (3.6)$$

In order for M_1 to in fact accelerate downward (that is, $a < 0$), we must have $M_2 < M_1(\sin \theta - \mu \cos \theta)$. Therefore, the range of M_2 for which the system doesn't accelerate (that is, it just sits there, assuming that it started at rest) is

$$M_1(\sin \theta - \mu \cos \theta) \leq M_2 \leq M_1(\sin \theta + \mu \cos \theta). \quad (3.7)$$

If μ is very small, then M_2 must essentially be equal to $M_1 \sin \theta$ if the system is to be static. Equation (3.7) also implies that if $\tan \theta \leq \mu$, then M_1 won't slide down, even if $M_2 = 0$. ♣

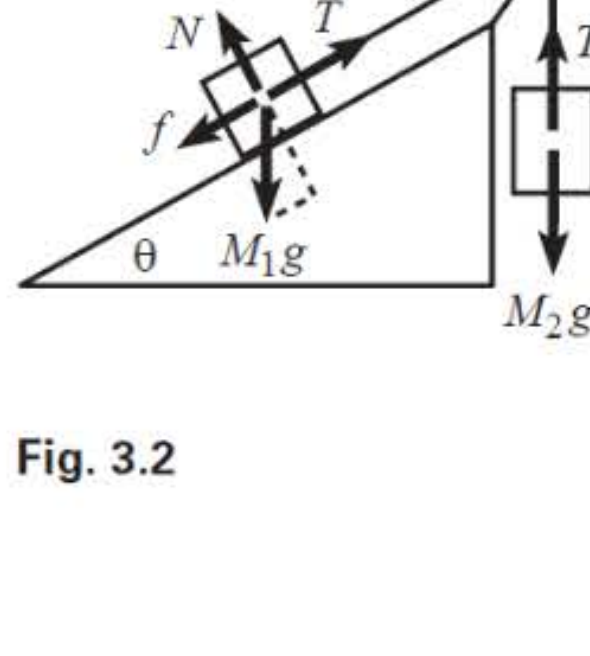


Fig. 3.2

- Let's apply $F = ma$ to the person. The forces acting on the person are gravity, the normal force from the platform, and the tension from the rope (pulling downward on her hand). So we have

$$N - T - mg = ma. \quad (3.9)$$

- Now apply $F = ma$ to the platform. The forces acting on the platform are gravity, the normal force from the person, and the force upward from the rod. So we have

$$f - N - Mg = Ma. \quad (3.10)$$

- Now apply $F = ma$ to the pulley. The forces acting on the pulley are gravity, the force downward from the rod, and twice the tension in the rope (because it pulls up on both sides). So we have

$$2T - f - \mu g = \mu a. \quad (3.11)$$

Note that if we add up the three previous equations, we obtain the $F = ma$ equation in Eq. (3.8), as should be the case, because the whole system is the sum of the three above subsystems. Equations (3.9)–(3.11) are three equations in the three unknowns, T , N , and f . Their sum yields the T in (3.8), and then Eqs. (3.9) and (3.11) give, respectively, as you can show,

$$N = (M + 2m + \mu)(g + a), \quad \text{and} \quad f = (2M + 2m + \mu)(g + a). \quad (3.12)$$

REMARKS: You can also obtain these results by considering subsystems different from the ones we chose above. For example, you can choose the pulley-plus-platform subsystem, etc. But no matter how you choose to break up the system, you will need to produce three independent $F = ma$ equations in order to solve for the three unknowns, T , N , and f .

In problems like this one, it's easy to forget to include certain forces, such as the second T in Eq. (3.11). The safest thing to do is to always isolate each subsystem, draw a box around it, and then draw all the forces that "protrude" from the box. In other words, draw the free-body diagram. Figure 3.4 shows the free-body diagram for the subsystem consisting of only the pulley. ♣

Problem 2)

Solution: To find the tension in the rope, we simply want to let our subsystem be the whole system (except the ceiling). If we imagine putting the system in a black box (to emphasize the fact that we don't care about any internal forces within the system), then the forces we see "protruding" from the box are the three weights (Mg , mg , and μg) downward, and the tension T upward. Applying $F = ma$ to the whole system gives

$$T - (M + m + \mu)g = (M + m + \mu)a \implies T = (M + m + \mu)(g + a). \quad (3.8)$$

To find the normal force N between the person and the platform, and also the tension f in the rod connecting the pulley to the platform, it is not sufficient to consider the system as a whole. This is true because these forces are internal forces to this system, so they won't show up in any $F = ma$ equations (which involve only external forces to a system). So we must consider subsystems:

- Let's apply $F = ma$ to the person. The forces acting on the person are gravity, the normal force from the platform, and the tension from the rope (pulling downward on her hand). So we have

$$N - T - mg = ma. \quad (3.9)$$

- Now apply $F = ma$ to the platform. The forces acting on the platform are gravity, the normal force from the person, and the force upward from the rod. So we have

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- Now apply $F = ma$ to the pulley. The forces acting on the pulley are gravity, the force downward from the rod, and twice the tension in the rope (because it pulls up on both sides). So we have

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Problem 3)

Solution: The first thing to note is that the tension T is the same everywhere throughout the massless string, because otherwise there would be an infinite acceleration of some part of the string. It then follows that the tension in the short string connected to m_2 is $2T$. This is true because there must be zero net force on the massless right pulley, because otherwise it would have infinite acceleration. The $F = ma$

equations for the two masses are therefore (with upward taken to be positive)

$$\begin{aligned} T - m_1 g &= m_1 a_1, \\ 2T - m_2 g &= m_2 a_2. \end{aligned} \quad (3.13)$$

We now have two equations in the three unknowns, a_1 , a_2 , and T . So we need one more equation. This is the "conservation of string" fact, which relates a_1 and a_2 . If we imagine moving m_2 and the right pulley up a distance d , then a length $2d$ of string has disappeared from the two parts of the string touching the right pulley. This string has to go somewhere, so it ends up in the part of the string touching m_1 (see Fig. 3.6). Therefore, m_1 goes down by a distance $2d$. In other words, $y_1 = -2y_2$, where y_1 and y_2 are measured relative to the initial locations of the masses. Taking two time derivatives of this statement gives our desired relation between a_1 and a_2 ,

$$a_1 = -2a_2. \quad (3.14)$$

Combining this with Eq. (3.13), we can now solve for a_1 , a_2 , and T . The result is

$$a_1 = g \frac{2m_2 - 4m_1}{4m_1 + m_2}, \quad a_2 = g \frac{2m_1 - m_2}{4m_1 + m_2}, \quad T = \frac{3m_1 m_2 g}{4m_1 + m_2}. \quad (3.15)$$



Fig. 3.6

REMARKS: There are all sorts of limits and special cases that we can check here. A couple are: (1) If $m_2 = 2m_1$, then Eq. (3.15) gives $a_1 = a_2 = 0$, and $T = m_1 g$. Everything is at rest. (2) If $m_2 \gg m_1$, then Eq. (3.15) gives $a_1 = 2g$, $a_2 = -g$, and $T = 3m_1 g$. In this case, m_2 is essentially in free fall, while m_1 gets yanked up with acceleration $2g$. The value of T is exactly what is needed to make the net force on m_1 equal to $m_1(2g)$, because $T - m_1 g = 3m_1 g - m_1 g = m_1(2g)$. You can check the case where $m_1 \gg m_2$.

For the more general case where there are N masses instead of two, the "conservation of string" statement is a single equation that relates all N accelerations. It is most easily obtained by imagining moving $N - 1$ of the masses, each by an arbitrary amount, and then seeing what happens to the last mass. Note that these arbitrary motions undoubtedly do *not* correspond to the actual motions of the masses. This is fine; the single "conservation of string" equation has nothing to do with the $N F = ma$ equations. The combination of all $N + 1$ equations is needed to constrain the motions down to a unique set. ♣

Problem 4)

- (a) $F = ma$ gives $dv/dt = -g$. Multiplying by dt and integrating yields $v = -gt + A$, where A is a constant of integration.¹⁴ The initial condition $v(0) = 0$ gives $A = 0$. Therefore, $dy/dt = -gt$. Multiplying by dt and integrating yields $y = -gt^2/2 + B$. The initial condition $y(0) = h$ gives $B = h$. Therefore,

$$y = h - \frac{1}{2}gt^2. \quad (3.28)$$

- (b) $F = ma$ gives $v dv/dy = -g$. Separating variables and integrating yields $v^2/2 = -gy + C$. The initial condition $v(h) = 0$ gives $v^2/2 = -gy + gh$. Therefore, $v \equiv dy/dt = -\sqrt{2g(h - y)}$. We have chosen the negative square root because the particle is falling. Separating variables gives

$$\int \frac{dy}{\sqrt{h - y}} = -\sqrt{2g} \int dt. \quad (3.29)$$

This yields $2\sqrt{h - y} = \sqrt{2g}t$, where we have used the initial condition $y(0) = h$. Hence, $y = h - gt^2/2$, as in part (a). In part (b) here, we essentially derived conservation of energy, as we'll see in Chapter 5.

Problem 5)

Solution: For simplicity in future formulas, let's write the drag force as $F_d = -\beta v$ (otherwise we'd have a bunch of $1/m$'s floating around). Taking upward to be the positive y direction, the force on the ball is

$$F = -mg - \beta v. \quad (3.30)$$

Note that v is negative here, because the ball is falling, so the drag force points upward, as it should. Writing $F = m dv/dt$ and separating variables gives

$$\int_0^{v(t)} \frac{dv'}{g + \alpha v'} = - \int_0^t dt'. \quad (3.31)$$

Integration yields $\ln(1 + \alpha v/g) = -\alpha t$. Exponentiation then gives

$$v(t) = -\frac{g}{\alpha} (1 - e^{-\alpha t}). \quad (3.32)$$

Writing $dy/dt = v(t)$, and then separating variables and integrating to obtain $y(t)$, yields

$$\int_h^{y(t)} dy' = -\frac{g}{\alpha} \int_0^t (1 - e^{-\alpha t'}) dt'. \quad (3.33)$$

Therefore,

$$y(t) = h - \frac{g}{\alpha} \left(t - \frac{1}{\alpha} (1 - e^{-\alpha t}) \right). \quad (3.34)$$

REMARKS:

1. Let's look at some limiting cases. If t is very small (more precisely, if $\alpha t \ll 1$), then we can use $e^{-x} \approx 1 - x + x^2/2$ to make approximations to leading order in t . You can show that Eq. (3.32) gives $v(t) \approx -gt$. This makes sense, because the drag force is negligible at the start, so the ball is essentially in freefall. And likewise you can show that Eq. (3.34) gives $y(t) \approx h - gt^2/2$, which is again the freefall result. We can also look at large t . In this case, $e^{-\alpha t}$ is essentially equal to zero, so Eq. (3.32) gives $v(t) \approx -g/\alpha$. (This is the "terminal velocity." Its value makes sense, because it is the velocity for which the total force, $-mg - \beta v$, vanishes.) And Eq. (3.34) gives $y(t) \approx h - (g/\alpha)t + g/\alpha^2$. Interestingly, we see that for large t , g/α^2 is the distance our ball lags behind another ball that started out already at the terminal velocity, $-g/\alpha$.
2. You might think that the velocity in Eq. (3.32) doesn't depend on m , since no m 's appear. However, there is an m hidden in α . The quantity α which we introduced just to make our formulas look a little nicer) was defined by $F_d = -\beta v \equiv -m\alpha v$. But the quantity $\beta = m\alpha$ is roughly proportional to the cross-sectional area, A , of the ball. Therefore, $\alpha \propto A/m$. Two balls of the same size, one made of lead and one made of styrofoam, have the same A but different m 's. So their α 's are different, and they fall at different rates. If we have a solid ball with density ρ and radius r , then $\alpha \propto A/m \propto r^2/(\rho r^3) = 1/\rho r$. For large dense objects in a thin medium such as air, the quantity α is small, so the drag effects are not very noticeable over short times (because if we include the next term in the expansion for v , we obtain $v(t) \approx -gt + \alpha g t^2/2$). Large dense objects therefore all fall at roughly the same rate, with an acceleration essentially equal to g . But if the air were much thicker, then all the α 's would be larger, and maybe it would have taken Galileo a bit longer to come to his conclusions.

What would you have thought, Galileo,
If instead you dropped cows and did say, "Oh!

Problem 6)

Let x and y be the horizontal and vertical positions, respectively. The force in the x direction is $F_x = 0$, and the force in the y direction is $F_y = -mg$. So $\mathbf{F} = m\mathbf{a}$ gives

$$\ddot{x} = 0, \quad \text{and} \quad \ddot{y} = -g. \quad (3.35)$$

Note that these two equations are "decoupled." That is, there is no mention of y in the equation for $\ddot{x}$, and vice versa. The motions in the x and y directions are therefore completely independent. The classic demonstration of the independence of the x and y motions is the following. Fire a bullet horizontally (or, preferably, just imagine firing a bullet horizontally), and at the same time drop a bullet from the height of the gun. Which bullet will hit the ground first? (Neglect air resistance, the curvature of the earth, etc.) The answer is that they will hit the ground at the same time, because the effect of gravity on the two y motions is exactly the same, independent of what is going on in the x direction.

If the initial position and velocity are (X, Y) and (V_x, V_y) , then we can easily integrate Eq. (3.35) to obtain

$$\dot{x}(t) = V_x, \quad \text{and} \quad \dot{y}(t) = V_y - gt. \quad (3.36)$$

Integrating again gives

$$x(t) = X + V_x t, \quad \text{and} \quad y(t) = Y + V_y t - \frac{1}{2}gt^2. \quad (3.37)$$

These equations for the speeds and positions are all you need to solve a projectile problem.

Problem 7)

- (a) Let the inclination angle be θ , and let the initial speed be V . Then the horizontal speed is always $V_x = V \cos \theta$, and the initial vertical speed is $V_y = V \sin \theta$. The first thing we need to do is find the time t in the air. We know that the vertical speed is zero at time $t/2$, because the ball is moving horizontally at the highest point. So the second of Eqs. (3.36) gives $V_y = g(t/2)$. Therefore, $t = 2V_y/g$.¹⁷ The first of Eqs. (3.37) tells us that the horizontal distance traveled is $d = V_x t$. Using $t = 2V_y/g$ in this gives

$$d = \frac{2V_x V_y}{g} = \frac{V^2(2 \sin \theta \cos \theta)}{g} = \frac{V^2 \sin 2\theta}{g}. \quad (3.38)$$

The $\sin 2\theta$ factor has a maximum at $\theta = \pi/4$. The maximum horizontal distance traveled is then $d_{\max} = V^2/g$.

REMARK: For $\theta = \pi/4$, you can show that the maximum height achieved is $V^2/4g$. This is half the maximum height of $V^2/2g$ (as you can show) if the ball is thrown straight up. Note that any possible distance you might want to find in this problem must be proportional to V^2/g , by dimensional analysis. The only question is what the numerical factor is. ♣

- (b) As in part (a), the first thing we need to do is find the time t in the air. If the ground is sloped at an angle β , then the equation for the line of the ground is $y = (\tan \beta)x$. The path of the ball is given in terms of t by

$$x = (V \cos \theta)t, \quad \text{and} \quad y = (V \sin \theta)t - \frac{1}{2}gt^2, \quad (3.39)$$

where θ is the angle of the throw, as measured with respect to θ (because this is not the ground). We must solve for t that makes $y = (\tan \beta)x$, because this gives the time when the path of the ball intersects the line of the ground. Using Eq. (3.39), we find that $y = (\tan \beta)x$ when

$$t = \frac{2V}{g} (\sin \theta - \tan \beta \cos \theta). \quad (3.40)$$

(There is, of course, also the solution $t = 0$.) Plugging this into the expression for x in Eq. (3.39) gives

$$x = \frac{2V^2}{g} (\sin \theta \cos \theta - \tan \beta \cos^2 \theta). \quad (3.41)$$

We must now maximize this value for x , which is equivalent to maximizing the distance along the slope. Setting the derivative with respect to θ equal to zero, and using the double-angle formulas, $\sin 2\theta = 2 \sin \theta \cos \theta$ and $\cos 2\theta = \cos^2 \theta - \sin^2 \theta$, we find $\tan \beta = -\cot 2\theta$. This can be rewritten as $\tan \beta = -\tan(\pi/2 - 2\theta)$. Therefore, $\beta = -(\pi/2 - 2\theta)$, so we have

$$\theta = \frac{1}{2} \left(\beta + \frac{\pi}{2} \right). \quad (3.42)$$

In other words, the throwing angle should bisect the angle between the ground and the vertical.

REMARKS:

1. For $\beta \approx \pi/2$, we have $\theta \approx \pi/2$, as should be the case. For $\beta = 0$, we have $\theta = \pi/4$, as we found in part (a). And for $\beta \approx -\pi/2$, we have $\theta \approx 0$, which makes sense.
2. A quicker method of obtaining the time in Eq. (3.40) is the following. Consider the set of tilted axes parallel and perpendicular to the ground; let these be the x' and y' axes, respectively. The initial velocity in the y' direction is $V \sin(\theta - \beta)$, and the acceleration in this direction is $g \cos \beta$. The time in the air is twice the time it takes the ball to reach the maximum "height" above the ground (measured in the y' direction), which occurs when the velocity in the y' direction is instantaneously zero. The total time is therefore $2V \sin(\theta - \beta)/(g \cos \beta)$, which you can show is equivalent to the time in Eq. (3.40). Note that the $g \sin \beta$ acceleration in the x' direction is irrelevant in calculating this time. In the present example, using these tilted axes doesn't save a huge amount of time, but in some situations (see Exercise 3.50) the tilted axes can save you a lot of grief.

3. An interesting fact about the motion of the ball in the maximum-distance case is that the initial and final velocities are perpendicular to each other. The demonstration of this is the task of Problem 3.16.
4. Substituting the value of θ from Eq. (3.42) into Eq. (3.41), you can show (after a bit of algebra) that the maximum distance traveled along the tilted ground is

$$d = \frac{x}{\cos \beta} = \frac{V^2/g}{1 + \sin \beta}. \quad (3.43)$$

Solving for V , we have $V^2 = g(d + d \sin \beta)$. This can be interpreted as saying that the minimum speed at which a ball must be thrown in order to pass over a wall of height h , at a distance L away on level ground, is given by $V^2 = g(\sqrt{L^2 + h^2} + h)$. This checks in the limits of $h \rightarrow 0$ and $L \rightarrow 0$.

5. A compilation of many other projectile results can be found in Buckmaster (1985). ♣

Problem 8)

- (a) $\mathbf{F} = m\mathbf{a}$ gives $\ddot{x} = -\alpha \dot{x}$ and $\ddot{y} = -g - \alpha \dot{y}$. Using the initial speed, the x equation integrates to

$$\dot{x} = A e^{-\alpha t} \implies x = v_0 \cos \theta e^{-\alpha t}. \quad (53)$$

Assuming an initial position of zero, this then integrates to

$$x = -(v_0 \cos \theta / \alpha) e^{-\alpha t} + B \implies x = (v_0 \cos \theta / \alpha) (1 - e^{-\alpha t}). \quad (54)$$

The y equation is $\ddot{y} = -\alpha(g/\alpha + \dot{y})$, which can be written as $(d/dt)(g/\alpha + \dot{y}) = -\alpha(g/\alpha + \dot{y})$. This integrates to

$$\frac{g}{\alpha} + \dot{y} = C e^{-\alpha t} \implies \dot{y} = C e^{-\alpha t} - \frac{g}{\alpha} \implies \dot{y} = \left(v_0 \sin \theta + \frac{g}{\alpha} \right) e^{-\alpha t} - \frac{g}{\alpha}. \quad (55)$$

This then integrates to

$$\begin{aligned} y &= -\frac{1}{\alpha} \left(v_0 \sin \theta + \frac{g}{\alpha} \right) e^{-\alpha t} - \frac{gt}{\alpha} + D \\ \implies y &= \frac{1}{\alpha} \left(v_0 \sin \theta + \frac{g}{\alpha} \right) (1 - e^{-\alpha t}) - \frac{gt}{\alpha}. \end{aligned} \quad (56)$$

- (b) We are given that $m\alpha v_0 = mg \implies g/\alpha = v_0$. Therefore, Eq. (55) gives $\dot{y} = (v_0 \sin \theta + v_0) e^{-\alpha t} - v_0$. At the top of the motion, we have $\dot{y} = 0 \implies e^{-\alpha t} = 1/(1 + \sin \theta)$. Using Eq. (54), the value of x at this time is</